

Planets Queries II

Time limit: 1.00 s

Memory limit: 512 MB

You are playing a game consisting of n planets. Each planet has a teleporter to another planet (or the planet itself).

You have to process q queries of the form: You are now on planet a and want to reach planet b . What is the minimum number of teleportations?

Input

The first input line contains two integers n and q : the number of planets and queries. The planets are numbered $1, 2, \dots, n$.

The second line contains n integers $t_1, t_2, \dots, t_n$: for each planet, the destination of the teleporter.

Finally, there are q lines describing the queries. Each line has two integers a and b : you are now on planet a and want to reach planet b .

Output

For each query, print the minimum number of teleportations. If it is not possible to reach the destination, print -1 .

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 3 2 3 2 3 2 1 2 1 3 1 4	1 2 -1